



Postdoctoral Researcher Position

Bio-inspired collective navigation in complex flows

Type: Theoretical/Numerical

Location: IRPHÉ, Marseille, France

Contract duration: up to 3 years, funded by the French National Research Agency (ANR)

Gross salary: between 3081 € and 4292 € per month, depending on experience

Starting date: flexible

Contact: Aurore Loisy (aurore.loisy@univ-amu.fr) <https://auroreloisy.github.io/>

Project. Inspired by the collective migration of numerous living organisms, this interdisciplinary project aims at uncovering the cooperation strategies that swarms can use to navigate and communicate in complex flows. The goal is to understand how hydrodynamical and chemical signals can be exploited at the group level for various tasks (e.g., foraging, efficient migration, ...), with applications to living systems and robotics. To this end, we will use a combination of theory, simulation and reinforcement learning. Depending on their background and preferences, the postdoctoral researcher will contribute to one or more of these aspects.

Profile. Candidates should have an outstanding track-record, some prior experience in either fluid mechanics (CFD simulations, transport/mixing in complex flows), active matter/biophysics or deep reinforcement learning, as well as a taste for interdisciplinary research and programming (Python).

Environment. The successful candidate will join the “Living and bio-inspired systems” group at the ‘Institut de recherche sur les phénomènes hors équilibre’ (IRPHÉ), a physics/mechanics lab located in Marseille. They will be part of an international team of researchers working on bio-inspired navigation, smart active particles and collective behaviour.

Application. Candidates are invited to contact Aurore Loisy by email. They should provide: (1) a detailed CV with publication list, (2) contact information of at least 2 academic references, and (3) a research statement (max 2 pages) describing past activities and research interests. Informal inquiries are also welcome.